APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

Black-Scholes-Merton Options Pricing

Model ID	APEX-MDL-0004
Version	1.0
Owner	Dr. Yuki Tanaka (Quant Researcher)
Validator	Dr. Samuel Achebe (Model Validation Officer)
Status	Validated
Last Review	2026-03-01
Next Review	2027-03-01

Regulatory Framework: SR 11-7, IFRS 13 Level 3

Confidential. Distribution restricted to authorised personnel only.
Educational and demonstration use only. Not connected to real financial infrastructure.

Model ID: APEX-MDL-0004
Status: Validated
Owner: Dr. Yuki Tanaka (Quant Researcher)
Validator: Dr. Samuel Achebe (Model Validation Officer)
Regulatory: SR 11-7, IFRS 13 Level 3

Contents

1 Model Overview **3**
1.1 Purpose . . . 3
1.2 Business Use . . . 3

2 Theoretical Basis **3**
2.1 Geometric Brownian Motion . . . 3
2.2 Black-Scholes Pricing Formulae . . . 3
2.3 Greeks . . . 4

3 Mathematical Specification **4**

4 Implementation **4**

5 Validation **4**

6 Model Limitations **5**

7 Open Findings **5**

1 Model Overview

1.1 Purpose

The Black-Scholes-Merton (BSM) model prices European call and put options on equities and provides the full Greeks suite used for risk management and XVA exposure simulation.

1.2 Business Use

- **Trade pricing:** Mark-to-market for all equity option positions (AAPL_CALL_200).
- **Greeks:** Delta/gamma/vega/theta/rho/charm fed to VaR Monte Carlo and FRTB-SA sensitivities.
- **XVA:** Delta and gamma used in PFE exposure simulation.
- **IFRS 13:** Deep OTM options classified Level 3 (unobservable vol inputs).

2 Theoretical Basis

2.1 Geometric Brownian Motion

Under the risk-neutral measure $\mathbb{Q}$, the stock price S follows:

$$dS = (r - q)\, S\, dt + \sigma S\, dW_t$$

with risk-free rate r, continuous dividend yield q, constant volatility σ, and Brownian motion W_t.

2.2 Black-Scholes Pricing Formulae

$$C = S\, e^{-qT} N(d_1) - K\, e^{-rT} N(d_2)$$
$$P = K\, e^{-rT} N(-d_2) - S\, e^{-qT} N(-d_1)$$

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

where $N(\cdot)$ is the standard normal CDF.

2.3 Greeks

Greek	Formula
Delta (call)	$e^{-qT} N(d_1)$
Delta (put)	$-e^{-qT} N(-d_1)$
Gamma	$\dfrac{e^{-qT}\,\phi(d_1)}{S\,\sigma\sqrt{T}}$
Vega	$S\,e^{-qT}\,\phi(d_1)\sqrt{T}$
Theta (call)	$-\dfrac{S\,\sigma\,e^{-qT}\,\phi(d_1)}{2\sqrt{T}} - r\,K\,e^{-rT} N(d_2) + q\,S\,e^{-qT} N(d_1)$
Rho (call)	$K\,T\,e^{-rT} N(d_2)$
Charm	$e^{-qT}\left[\phi(d_1)\frac{2(r-q)T - d_2\,\sigma\sqrt{T}}{2T\,\sigma\sqrt{T}} - q\,N(d_1)\right]$

$\phi(\cdot)$ denotes the standard normal PDF.

3 Mathematical Specification

Parameter	Value
Risk-free rate r	4.5% (US OIS approximation)
Dividend yield q	0% (no dividends modelled)
Implied volatility σ	30% (flat surface, calibrated quarterly)
Time to expiry T	0.25 years (90-day standard)
Option multiplier	100 shares/contract

4 Implementation

Code location: `infrastructure/trading/greeks.py` — `GreeksCalculator._option_greeks()`

Key implementation notes:

- `_CALL_` in ticker ⇒ call option; `_PUT_` ⇒ put option.
- Multiplier 100 applied to all Greeks.
- `scipy.stats.norm` used for $N(\cdot)$ and $\phi(\cdot)$.
- $T = 0.25$ hardcoded (review quarterly).

5 Validation

Benchmark: Call prices validated against put-call parity ($C - P = Se^{-qT} - Ke^{-rT}$); max error $< \$0.01$.

Greeks backtesting: Delta P&L explain: $\Delta P \approx \Delta\,\Delta S + \frac{1}{2}\Gamma(\Delta S)^2$; $R^2 > 0.98$ for daily hedges.

IFRS 13 assessment: ATM options (within ±10% moneyness) classified Level 2. Deep OTM/ITM (>20% moneyness) classified Level 3 per IFRS 13.81.

6 Model Limitations

1. **Flat volatility surface:** No smile or skew; BSM underprices OTM puts and overprices OTM calls relative to market.
2. **Constant volatility:** Stochastic vol (Heston, SABR) not implemented; BSM vega risk may be misestimated by ±30% for long-dated options.
3. **No early exercise:** European-style only; American options mispriced.
4. **No discrete dividends:** Continuous dividend yield is a poor approximation for discrete ex-dividend events.
5. **Fixed $T = 0.25$:** Fails to reflect time decay correctly for options approaching expiry or beyond 90 days.

7 Open Findings

ID	Severity	Description	Status
BSM-F1	**Major**	Flat volatility surface — no smile/skew calibration; material mispricing for OTM options (>15% moneyness).	Open
BSM-F2	**Minor**	No discrete dividend handling; continuous yield approximation used; error <2% for low-yield stocks.	Open